

[Internet of Things]
[(Open Ended Lab) Report]



[Smart Door Automation System]

Submitted by:

Mudassar Hussain	23L-6006
Waleed Maqbool	23L-6012

Submitted to:

Miss Sughra Kamran

[7/05/2026]

Department of Electrical Engineering
National University of Computer and Emerging Sciences, Lahore

Table of Contents

1	Introduction
2	Problem Analysis
3	Design Requirements
4	Feasibility Analysis
5	Possible Solutions
6	Preliminary Design
7	Design Description
8	Software Simulation
9	Experimental Results
10	Performance Analysis
11	Conclusion

References

Introduction

With the rapid advancement in embedded systems and automation technologies, smart systems are increasingly being used to improve efficiency, safety, and real-time decision-making. This project focuses on the design and implementation of a **Smart Door Automation System** that operates based on environmental conditions such as temperature and humidity.

The **core idea** of this system is to continuously monitor ambient conditions using a temperature and humidity sensor (DHT11/DHT22) and automatically control a door mechanism (simulated using an LED). When the environmental parameters exceed predefined threshold values, the system activates the LED, representing the opening of a door. Otherwise, the system keeps the door closed. This automation reduces the need for human intervention and ensures timely response to changing environmental conditions.

Furthermore, this project introduces the concept of **automation based on environmental intelligence**, which is a foundational idea in modern Internet of Things (IoT) systems. Although this implementation uses a simple LED as an actuator, the same concept can be extended to control real-world devices such as motors, fans, or automated doors.

Problem Analysis

The problem presented in this project is to design a system that can **monitor environmental conditions (temperature and humidity)** and automatically **control a door mechanism** based on those conditions. The system must continuously observe real-time data and take appropriate action when certain limits are exceeded. Additionally, it should provide feedback to the user by displaying sensor readings.

In many real-world environments such as industrial facilities, storage areas, and controlled rooms, environmental conditions play a critical role in safety and operational efficiency. Manually monitoring these conditions is not only inefficient but also prone to delays and human error. Therefore, the key challenge is to develop a system that can **automatically detect changes and respond instantly without human intervention**.

The project specifically requires:

- Interfacing a temperature and humidity sensor
- Defining threshold values for environmental conditions
- Making decisions based on sensor readings
- Controlling an output device (LED) to simulate a door mechanism
- Displaying real-time data

To solve this problem, an **embedded system-based approach** is used. A DHT11 sensor is integrated with a microcontroller (ESP8266) to collect real-time temperature and humidity data. The microcontroller processes this data and compares it with predefined threshold values. If either the temperature or humidity exceeds the set limits, the system activates the LED, representing the opening of the door. If the conditions remain within safe limits, the LED stays OFF, indicating that the door remains closed.

Design Requirements

The design requirements of the Smart Door Automation System are based on the need to monitor environmental conditions and control a door mechanism accordingly. The system must operate reliably, respond in real-time, and remain simple and cost-effective.

Input Requirements

The system takes environmental data as input through a temperature and humidity sensor.

- **Temperature Input:** Measured in degrees Celsius using the DHT11 sensor
- **Humidity Input:** Measured as a percentage (%) using the same sensor

These inputs are continuously acquired in real time and sent to the microcontroller for processing.

Output Requirements

The system produces outputs based on the processed input data.

- **LED Output:**
 - ON → Indicates door is open (threshold exceeded)
 - OFF → Indicates door is closed (normal conditions)
- **Serial Monitor Output:**
 - Displays real-time temperature and humidity readings
 - Shows system status messages (e.g., "Threshold Exceeded" or "Normal")

Functional Requirements

- Continuously read temperature and humidity values
- Compare sensor readings with predefined threshold values

Activate LED when:

- Temperature > 30°C **OR**
- Humidity > 40%
- Keep LED OFF when conditions are within safe limits

Design Requirements

Non-Functional Requirements

- **Real-Time Operation:** System should update readings every 2 seconds
- **Reliability:** Must handle sensor failures safely (LED OFF in error case)
- **Simplicity:** Easy to implement and understand
- **Low Cost:** Use affordable components like DHT11 and ESP8266

Constraints

Sensor Limitations:

- DHT11 has limited accuracy ($\pm 2^{\circ}\text{C}$ temperature, $\pm 5\%$ humidity)
- Slow sampling rate (~ 2 seconds)

Hardware Constraints:

- Limited GPIO pins on microcontroller
- Power supply limitations

Environmental Constraints:

- Sensor performance may vary with extreme conditions

Feasibility Analysis

Time Feasibility

The project is highly feasible within the given time constraint of **1 hour** due to its simple design and implementation.

- The system uses **basic sensor interfacing**, which is straightforward and already supported by existing libraries (DHT library).
- The code logic is simple, involving reading sensor data, comparing thresholds, and controlling an LED.

Time Breakdown (Approximate)

- Circuit setup: 10–15 minutes
- Code writing/uploading: 15–20 minutes
- Testing and debugging: 15–20 minutes

Cost Feasibility

The project is **highly cost-effective** and feasible under the constraint of **no additional cost**.

Required Components (Lab Equipment):

- ESP8266 / ESP32
- DHT11 sensor
- LED and resistor
- Breadboard and jumper wires

Resource Management

- Efficient use of available lab equipment
- Minimal hardware requirements
- Reusable components for future projects

Possible Solutions

Solution 1: Manual Monitoring System

In this approach, a user manually observes temperature and humidity readings and decides whether to open or close the door.

Advantages:

- No hardware or programming required
- Simple implementation

Disadvantages:

- Not real-time
- High chances of human error

Solution 2: Microcontroller-Based Automated System (Chosen Solution)

This approach uses a microcontroller (ESP8266/Arduino) with a DHT sensor to automatically control an LED (door simulation) based on predefined thresholds.

Advantages:

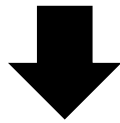
- Fully automated system
- Real-time monitoring and response
- Low cost and easy implementation
- Reliable and efficient

Disadvantages:

- Limited to local control (no remote access)
- Depends on sensor accuracy

Preliminary Design

DHT 11 Sensor
(Temp & Humidity)



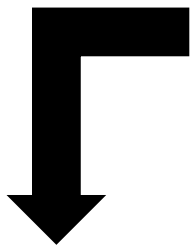
ESP 8266/ESP 32



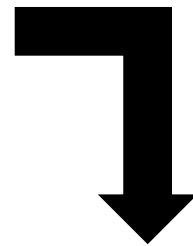
Decision Logic



Output



Serial Monitor



Led (Door)

Design Description

1. Sensor (DHT11 Temperature & Humidity Sensor)

The sensor block is responsible for collecting environmental data.

- The DHT11 sensor measures:
 - **Temperature (°C)**
 - **Humidity (%)**
- It provides **digital output**, which simplifies interfacing with the microcontroller.
- The sensor operates with a sampling interval of approximately **2 seconds**.
- It sends real-time data to the microcontroller through a single data pin.

Role in System:

Acts as the input source by continuously monitoring environmental conditions.

2. Processing (Microcontroller – ESP8266 / ESP32)

The processing block is the core of the system where all decisions are made.

- Initializes sensor and communication (Serial Monitor)
- Reads temperature and humidity values from the sensor
- Checks for invalid readings using error handling (isnan() function)
- Executes programmed logic to determine system behavior

Role in System:

Processes incoming data and controls overall system operation.

3. Decision Logic

This block implements the condition-based control mechanism.

- Predefined threshold values:
 - Temperature Threshold = **30°C**
 - Humidity Threshold = **40%**
- The system evaluates the condition:
 - If **temperature > threshold OR humidity > threshold**
→ Trigger action

Design Description

- Otherwise → Maintain normal state

4. Output (LED – Door Simulation)

The output block provides a visual indication of system status.

- **LED ON:** Door is considered **OPEN** (threshold exceeded)
- **LED OFF:** Door is **CLOSED** (normal conditions)

Role in System:

Represents the actuation mechanism (door control) based on decision logic.

5. Display/Monitoring (Serial Monitor)

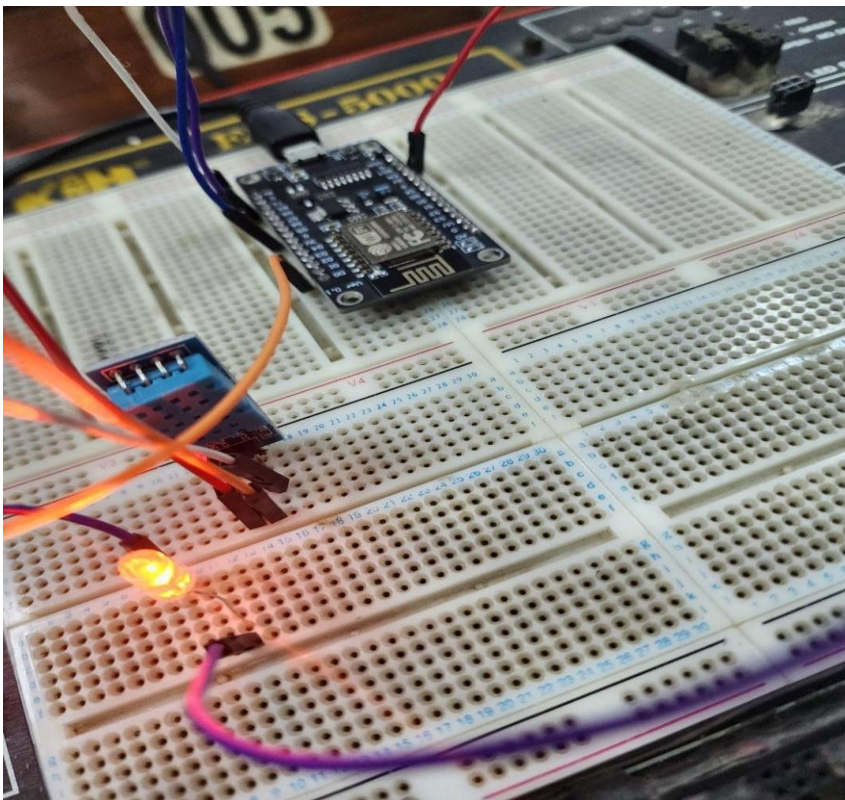
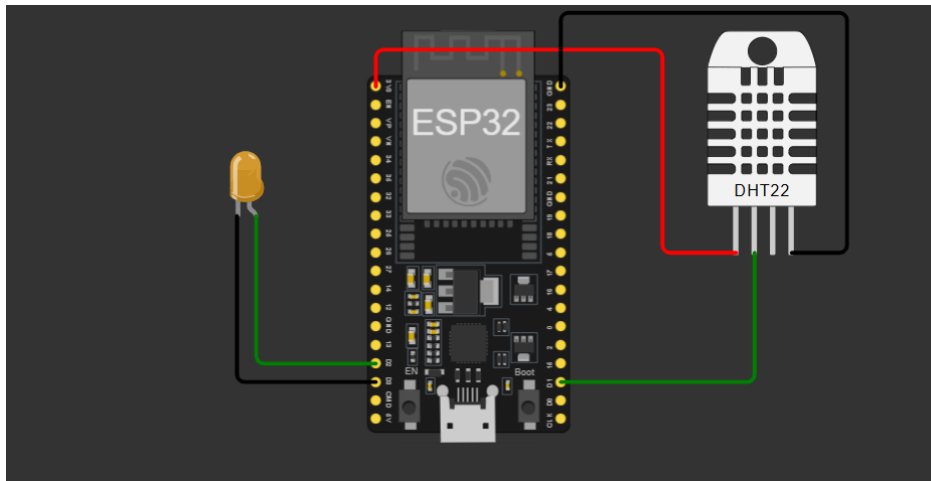
This block is used for real-time observation and debugging.

- Displays:
 - Temperature values
 - Humidity values
 - System status messages (e.g., “LED ON”, “Normal”)

Role in System:

Provides user feedback and system transparency.

Software Simulation (Optional)



Experimental Results

Observed Results

During testing, temperature and humidity values were read in real-time and displayed on the Serial Monitor. The LED responded according to the defined threshold conditions.

Temperature (°C)	Humidity (%)	LED Status (Door)	System Response
27	35	OFF	Normal
29	38	OFF	Normal
31	41	ON	Threshold Exceeded

Serial Monitor:

```
Temp: 23.80C | Hum: 41.00%  
Threshold Exceeded: LED ON  
-----  
Temp: 23.80C | Hum: 41.00%  
Threshold Exceeded: LED ON  
-----  
Temp: 23.80C | Hum: 41.00%  
Threshold Exceeded: LED ON  
-----  
Temp: 23.80C | Hum: 41.00%  
Threshold Exceeded: LED ON  
-----  
Temp: 23.80C | Hum: 41.00%  
Threshold Exceeded: LED ON  
-----
```

Key Observations

- The system continuously reads sensor data every 2 seconds.
- LED turns ON when:
 - Temperature exceeds 30°C
 - OR Humidity exceeds 40%
- LED remains OFF when both values are within safe limits.

Performance Analysis

1. Accuracy Analysis

The system shows **acceptable accuracy** for basic environmental monitoring.

- The DHT11 sensor provides approximate readings with:
 - $\pm 2^{\circ}\text{C}$ accuracy for temperature
 - $\pm 5\%$ accuracy for humidity
- The system correctly triggers the LED when threshold values are exceeded.

Analysis:

Although the sensor is not highly precise, it is sufficient for this application. Minor variations in readings do not significantly affect system behavior since the logic is threshold-based.

2. Response Time

- The system updates readings every **2 seconds**, which matches the DHT11 sampling rate.
- The LED response is almost **instant after data is read**.

Analysis:

The response time is suitable for real-time monitoring in small-scale systems. However, it may not be ideal for applications requiring ultra-fast response.

3. Reliability

- The system includes error handling using `isnan()` to detect invalid sensor readings.
- In case of failure, the LED remains OFF, preventing false alerts.

Analysis:

This makes the system **safe and reliable**, as it avoids incorrect actuation due to faulty data.

4. System Stability

- Continuous operation in loop without crashes
- Consistent output during repeated testing
- No unexpected fluctuations in LED behavior

Analysis:

The system demonstrates stable performance under normal conditions.

Conclusion

The Smart Door Automation System was successfully designed and implemented using a DHT11 temperature and humidity sensor with a microcontroller-based control unit. The system effectively monitors environmental conditions in real time and responds by activating an LED that simulates a door mechanism.

Based on the experimental results and performance analysis, it is concluded that the system fulfills all the defined objectives of the project. The sensor accurately captures environmental data, and the microcontroller processes this data correctly using predefined threshold conditions. Whenever the temperature or humidity exceeds the set limits, the system reliably turns ON the LED, indicating an automatic response, while maintaining the LED OFF state under normal conditions.

In conclusion, the developed solution is practical, cost-effective, and suitable for educational purposes. It effectively demonstrates how environmental data can be used to control physical systems automatically, forming the basis for more advanced smart systems and IoT-based applications in the future.

References

- [1] ESP8266/ESP32 NodeMCU Documentation
(Used for microcontroller interfacing and pin configuration)
- [2] Embedded Systems and IoT Fundamentals (Course Material)
(Used for concepts of sensor interfacing, data acquisition, and automation logic)
- [3] Internet of Things (IoT) Concepts
(Used for understanding automation systems and real-time monitoring concepts)
- [4] YouTube

Range of Complex Engineering Activity

EA1: Range of resources EA2: Level of interaction EA3: Innovation EA4: Consequences for society and environment EA5: Familiarity	• EA1: Range of resources -- The design involves the use of diverse resources, such as, money, equipment, information, and technology. • EA2: Level of Interaction – The design requires resolution of various problems arising from interactions between wide-ranging or conflicting technical or other issues. • EA3: Innovation – The design addresses sustainability and reduces cost requiring creative use of engineering principles and research-based knowledge in novel ways. • EA5: Familiarity – The design activity emphasizes the integration of existing knowledge and tools (or familiar solutions) with new or unfamiliar challenges			
	Rubrics		LLOs	Marks
	Explains the design process including engagement of resources clearly	EA1	LLO3	
	Demonstrate the final design clearly with all supporting information	EA1, EA5	LLO4	
	Demonstrate how innovation has been used in design to resolve conflicting requirements including the impact on society and environment	EA2, EA3	LLO4	
	Prepares a report which explain the design process including engagement of resources clearly, and is free from grammatical errors.	EA1	LLO3	

